Patient-specific balloon design for angioplasty by shape optimization

Thesis advisors:

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SPHINX, IECL, Nancy : Yannick Privat



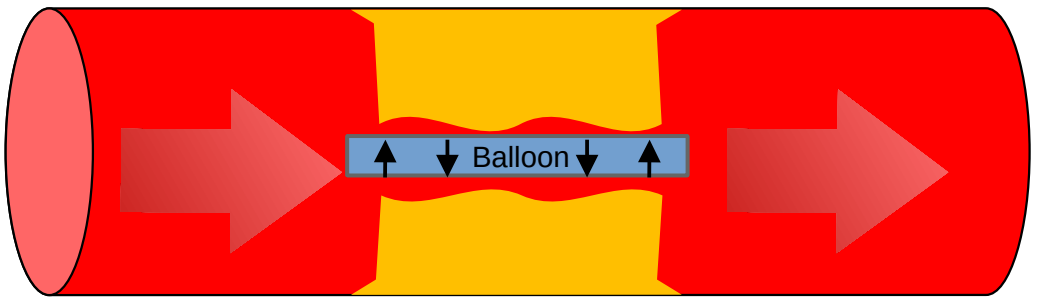




Context – Patient specific balloon design.

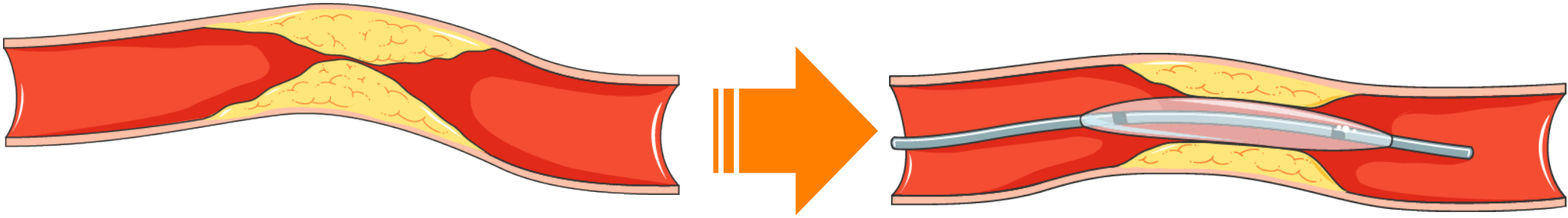
Deposit on the artery wall: Stenosis[1].

A known treatment: Angioplasty (balloon)[1].



[1] [Martin et al., 2010], [Morlacchi et al., 2013], [Pan et al., 2021].

Context – Patient specific balloon design.

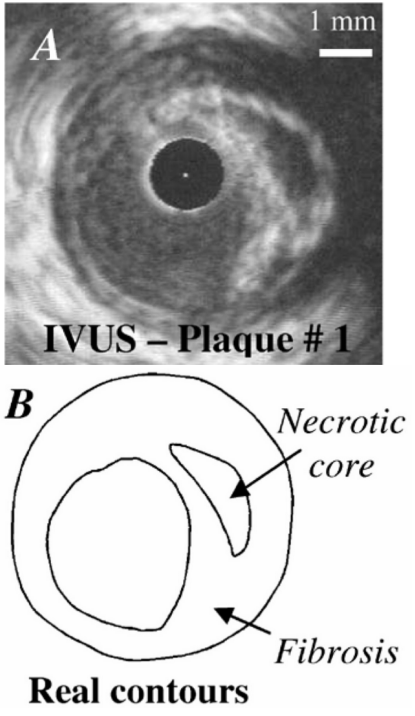


Current approach: Symmetric balloon (cylinder)[1].

Problem: the stenosis is not symmetric, it depends on the patient

(geometry of the artery, properties of the plaque of lipids, heterogeneity of the arterial wall...)

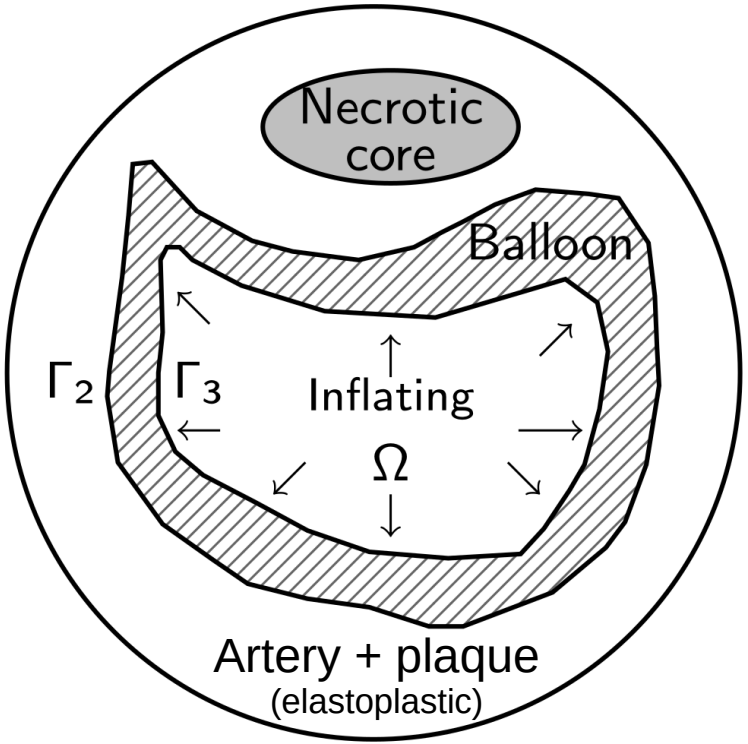
Objective: Designing balloons that are better tailored to patient.



[1] [Antonini et al., 2023].

Picture: [Le Floc'h et al., 2009].

Balloon problem: modeling.



Goal: Ease the blood flow by adapting the thickness of the balloon.

Our approach: Optimize Γ_3 such that after inflating and deflating the balloon, $\Gamma_2 \approx$ circle using shape optimization via level-set method[1].

[1] [Osher, Sethian, 1988].

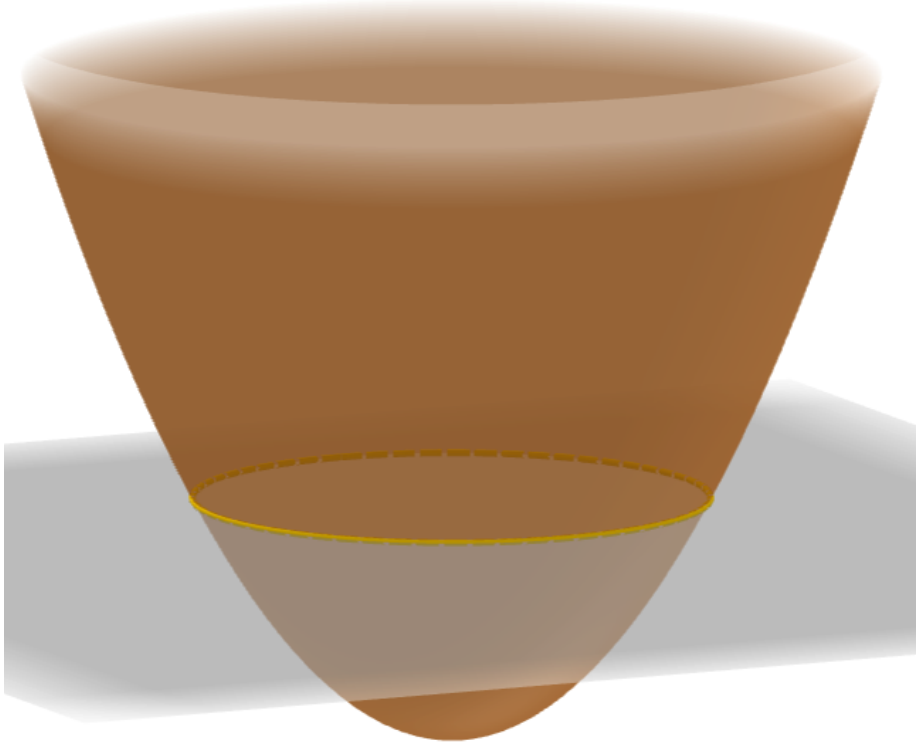
Level-set algorithm.[1]

Ansatz: $\Omega = \{\varphi < 0\}$.

Goal: If J is a cost function, calculate

$$\Omega = \underset{\Omega}{\operatorname{argmin}} J(\Omega).$$

Algorithm: Gradient descent.

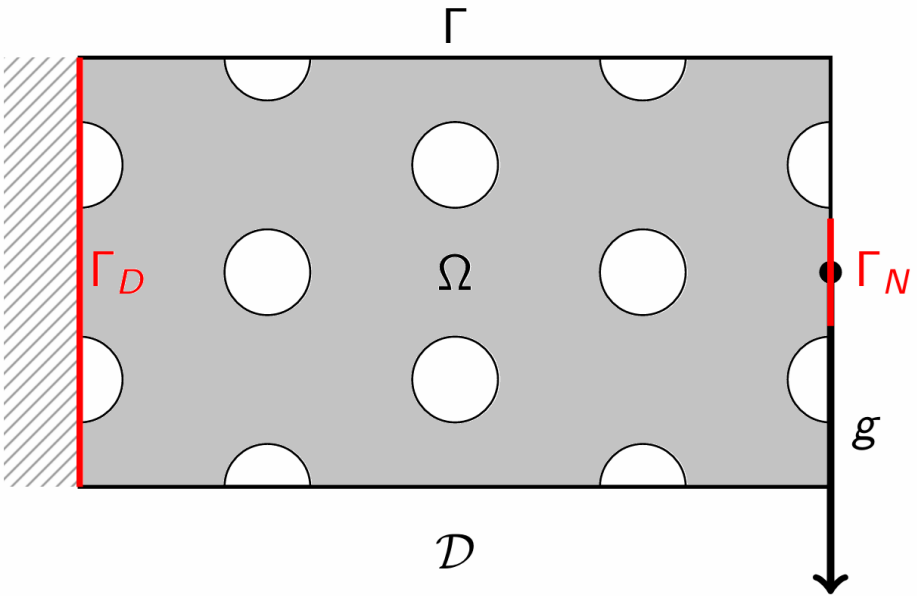


Steps:

1) Computation of $J(\Omega)$: state equations.
2) Computation of the shape derivative.
3) Computation of an admissible perturbation.
4) Update Ω by updating φ.

[1] [Allaire et al., 2004].

Toy model: Cantilever beam.[1]



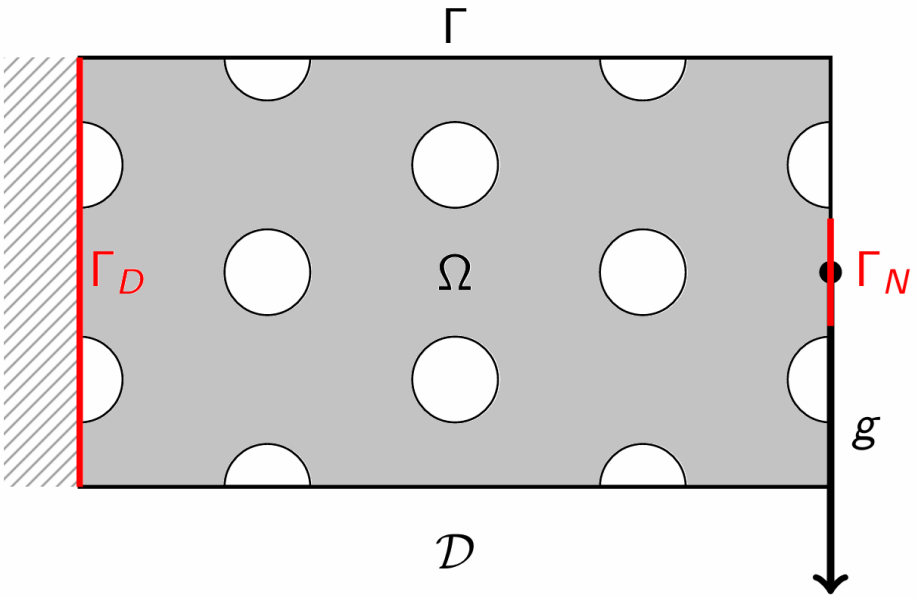
Goal:
For a fixed volume, find a structure Ω providing the best possible resistance when subjected to a force *g*.

Cost functional: Compliance.

[1] [Allaire et al., 2004], [Laurain, 2018].

Toy model: Cantilever beam.

Step 1: Computation of the cost function.



State equations:

$$
\begin{aligned}
-\mathrm{div}(\sigma(u)) &= 0 \quad \text{in } \Omega,\\
u &= 0 \quad \text{on } \Gamma_D,\\
\sigma(u)n &= g \quad \text{on } \Gamma_N,\\
\sigma(u)n &= 0 \quad \text{on } \Gamma,
\end{aligned}
$$

$$\sigma(u) = 2\mu\varepsilon(u) + \lambda\mathrm{div}(u)I.$$

Cost function:

$$J(\Omega) := \frac{1}{2}\int_\Omega \sigma(u) : \varepsilon(u)\mathrm{d}x.$$

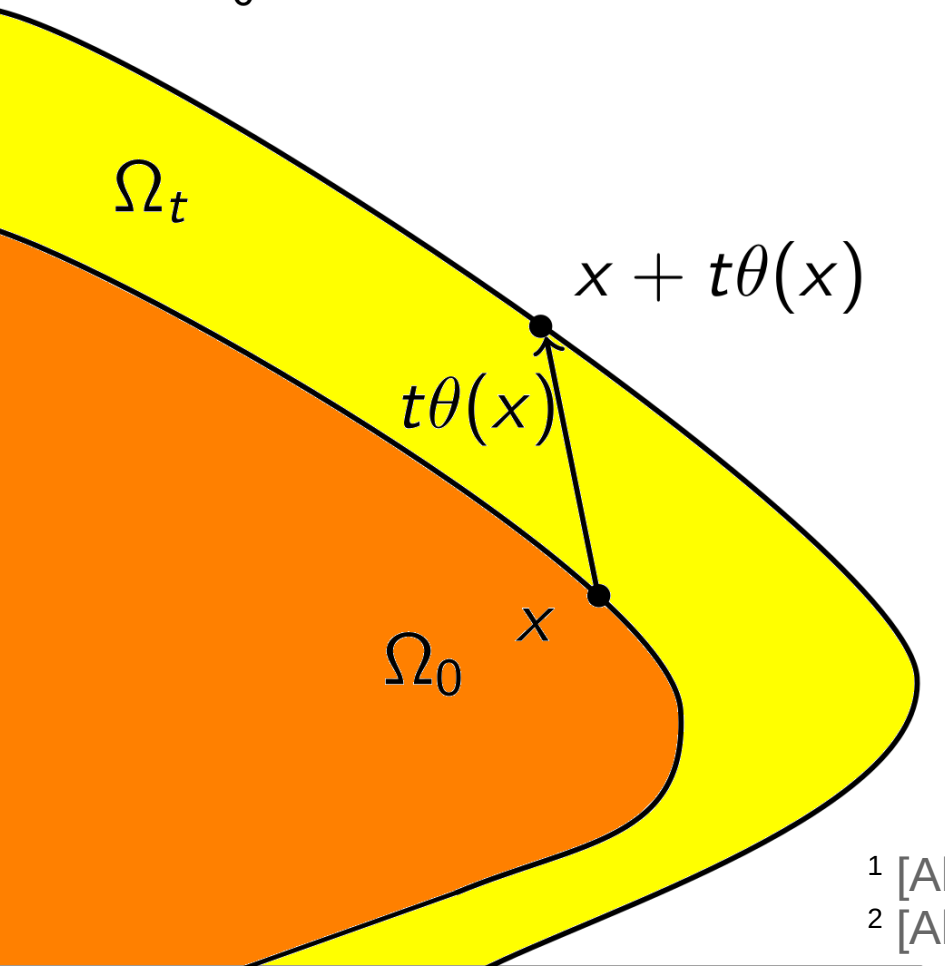
Numerical scheme: φ-FEM[1] (use of level-set).

[1] [Duprez et al., 2020].

Toy model: Cantilever beam.

Step 2: Shape derivative.[1]

Let Ω_0 be a bounded domain of $\mathbb{R}^N$ and θ a vector field (perturbation).



$$\mathrm{d}J_{\Omega_0}(\theta) := \lim_{t \to 0} \frac{J(\Omega_t) - J(\Omega_0)}{t}$$

$$\overset{2}{=} \frac{1}{2} \int_{\Omega_0} [2(\nabla u)^T \sigma(u) - \sigma(u) : \varepsilon(u) I] : \nabla\theta \, \mathrm{d}x.$$

[1] [Allaire, 2007], [Delfour, Zolésio, 2011].
[2] [Allaire et al., 2004], [Laurain, 2018].

Toy model: Cantilever beam.

Step 3: Computation of the perturbation.

Gradient descent:[1] for a given Ω, find θ (*perturbation*) such that:

$$\mathrm{d}J_\Omega(\theta) < 0$$

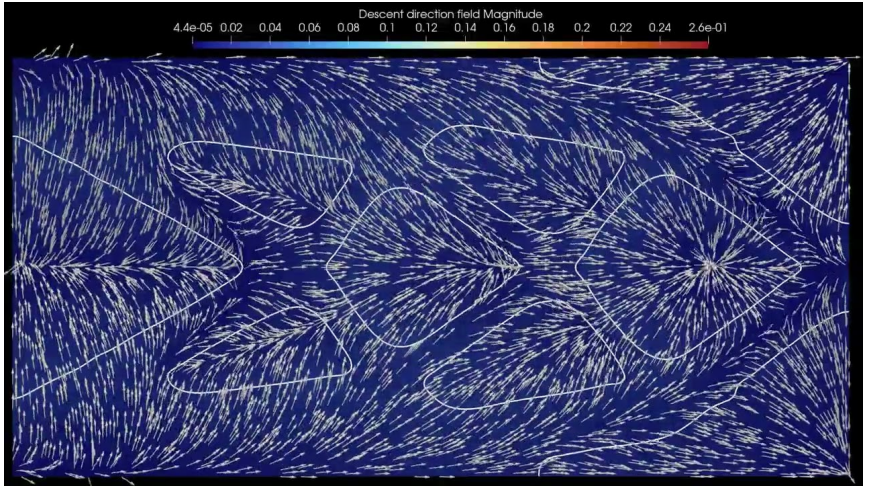
Find θ using a FE scheme s.t.:

$$\forall \xi, \quad \mathcal{B}(\theta, \xi) = -\mathrm{d}J_{\Omega_0}(\xi),$$

$\mathcal{B}$ bilinear, continuous, coercive.[2]

$$\mathrm{d}J_{\Omega_0}(\theta) = -\mathcal{B}(\theta, \theta) < 0$$

Calculation using finite elements (FE).

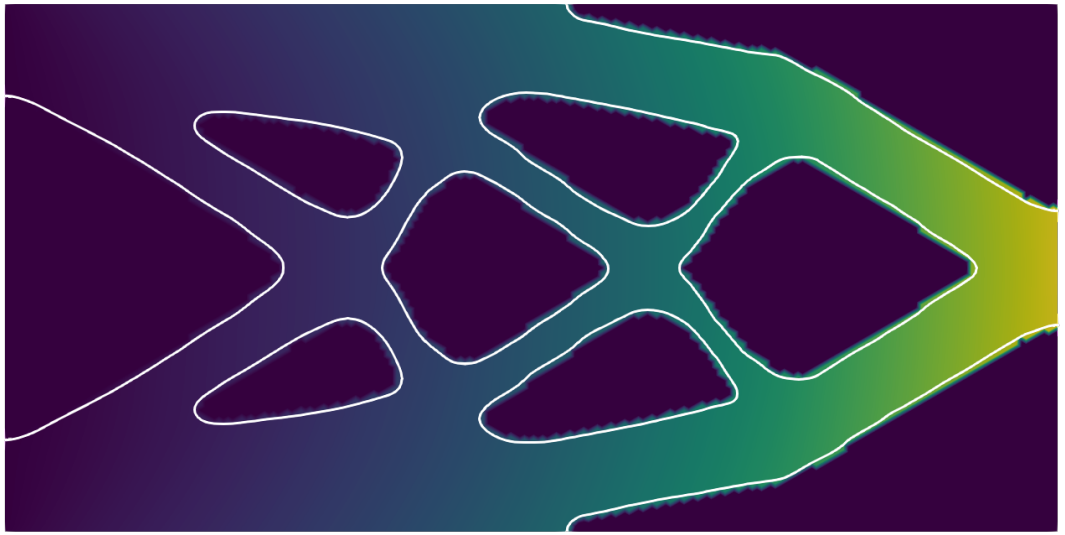


Video from R. Bulle.

[1] [Allaire et al., 2004].

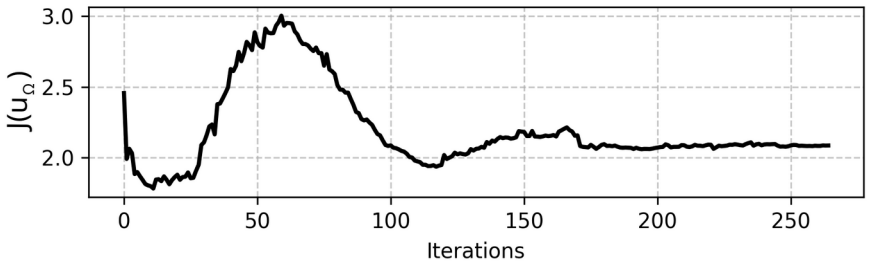
[2] [Burger, 2003].

Toy model: Cantilever beam. Result.[1]

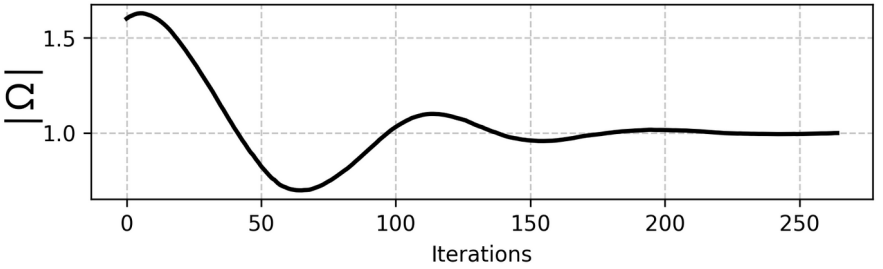


Video from R. Bulle.

Evolution of the cost function:



Evolution of the volume:



[1] Use of FormOpt: [Díaz-Avalos, Laurain, 2026]

AI perspectives.

In the shape optimization algorithm:

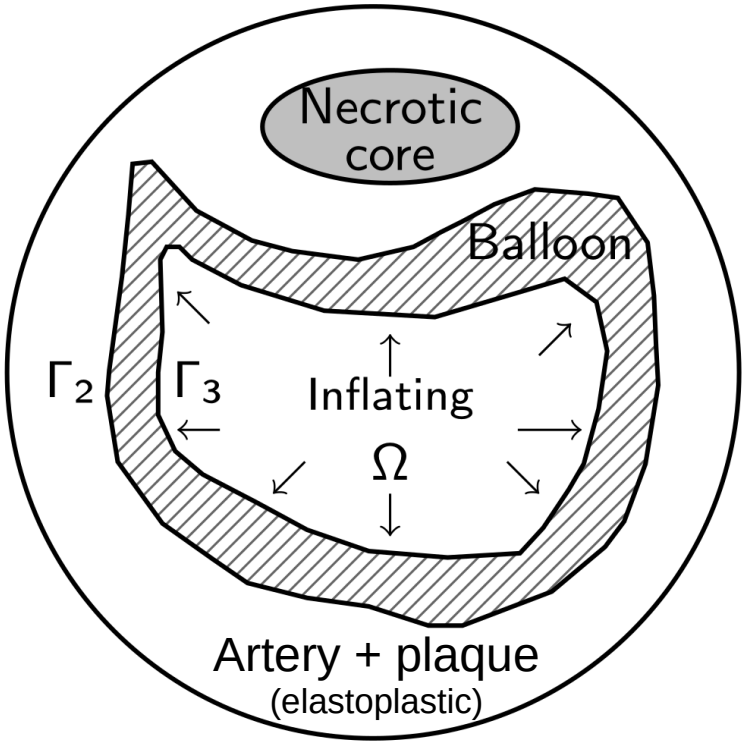
- States and perturbation: training a neural network (reducing cost).
  Existing work with φ-FEM-FNO: [Duprez et al., 2025].
  Neural network within the shape optimization loop: [Wang et al., 2024].
- Computation of $J(\Omega)$ (hard for the cantilever, worse for the balloon : more expensive, nonlinear equations…).
- Parameters of the optimization algorithm.
- Several at once?

In the balloon problem:

- Patient geometry as input.

Additional slides

Functional to minimize – Balloon problem.



Curvature variance:

$$J_0(\Gamma_2) = \frac{1}{2}\int_{\Gamma_2}\left(\kappa_\Omega - \frac{1}{|\Gamma_2|}\int_{\Gamma_2}\kappa_\Omega \mathrm{d}s\right)^2 \mathrm{d}s.$$

We minimize:

$$J : \Omega \longmapsto J_0((\mathrm{Id} + u_\Omega)(\Omega)).$$

Willmore functionals: see e.g. [Dogan, Nochetto, 2012].

Complete shape optimization algorithm.

1. Solving the primal (+ adjoint) problems: FE scheme.
2. Computation of the perturbation: Find θ using a FE scheme s.t.:

$$\forall \xi, \quad \mathcal{B}(\theta, \xi) = -\mathrm{d}J_{\Omega_0}(\xi).$$

Taking $\xi = \theta$ gives $\mathrm{d}J_{\Omega_0}(\theta) = -\mathcal{B}(\theta, \theta) < 0$ (Gradient descent).

3. Transport of the level-set: Find ϕ solution to:

$$\partial_t \phi + \theta \cdot \nabla \phi = 0, \text{ with } \phi(0, \cdot) = \varphi$$

[Osher, Sethian, 1988], [Burger, 2003], [Allaire et al., 2004].

Addition to the algorithm: the reinitialization.

4. Reinitialization of the level-set: Find ψ solution to:

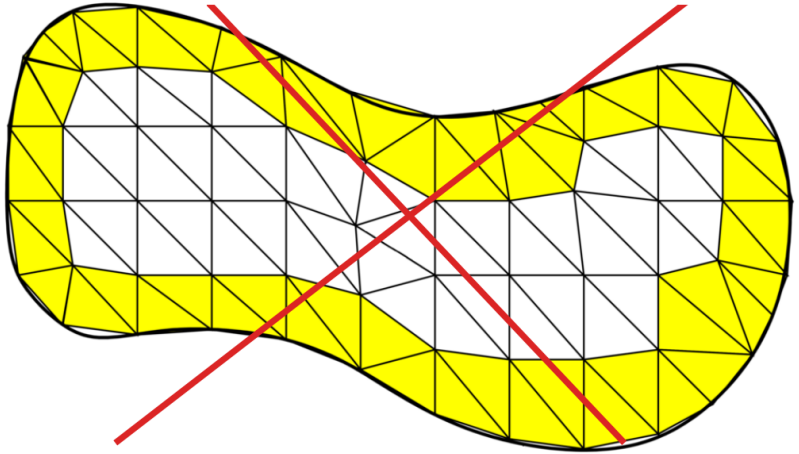
$$\partial_t \psi + \mathcal{S}(\phi)(|\nabla \psi| - 1) = 0, \text{ with } \psi(0, \cdot) = \phi.$$

$\Longrightarrow$ $|\nabla \varphi| \approx 1$.

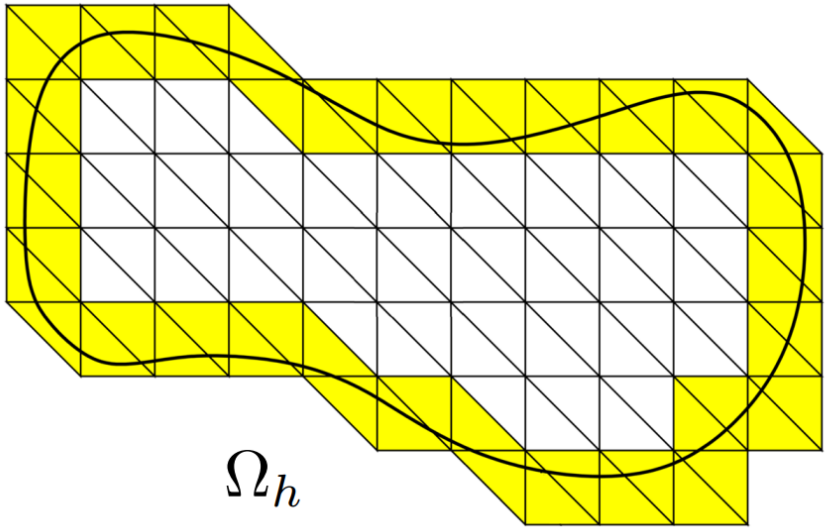
Goal: avoid the degeneration of the gradient of the level-set.

[Osher, Sethian, 1988], [Burger, 2003], [Allaire et al., 2004].

Immersed boundary methods.



Fitted mesh.

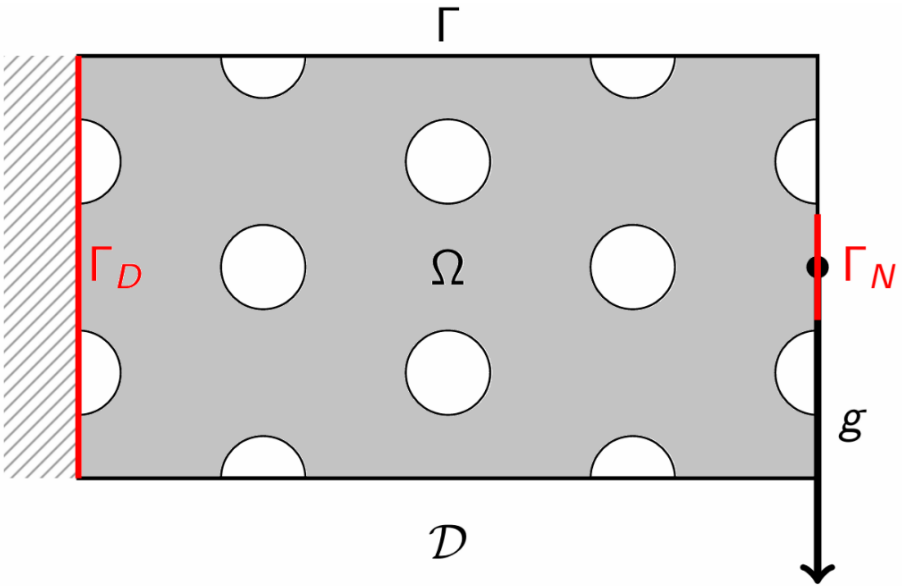


Unfitted mesh.

Remeshing at every iteration not necessary.
→ Suitable for shape optimization problems and the level-set method.

Our approach: φ-FEM [Duprez et al., 2020].
Other approaches: XFEM [Moes et al., 1999], CutFEM [Burman, Hansbo, 2014]…
Immersed boundary methods: [Saul'ev, 1963], [Glowinski et al., 1964a].

Cantilever beam: cost functional.



$$
\begin{aligned}
-\mathrm{div}(\sigma(u)) &= 0 \quad \text{in } \Omega,\\
u &= 0 \quad \text{on } \Gamma_D,\\
\sigma(u)n &= g \quad \text{on } \Gamma_N,\\
\sigma(u)n &= 0 \quad \text{on } \Gamma,
\end{aligned}
$$

$$\sigma(u) = 2\mu\varepsilon(u) + \lambda \mathrm{div}(u) I.$$

Optimization problem:

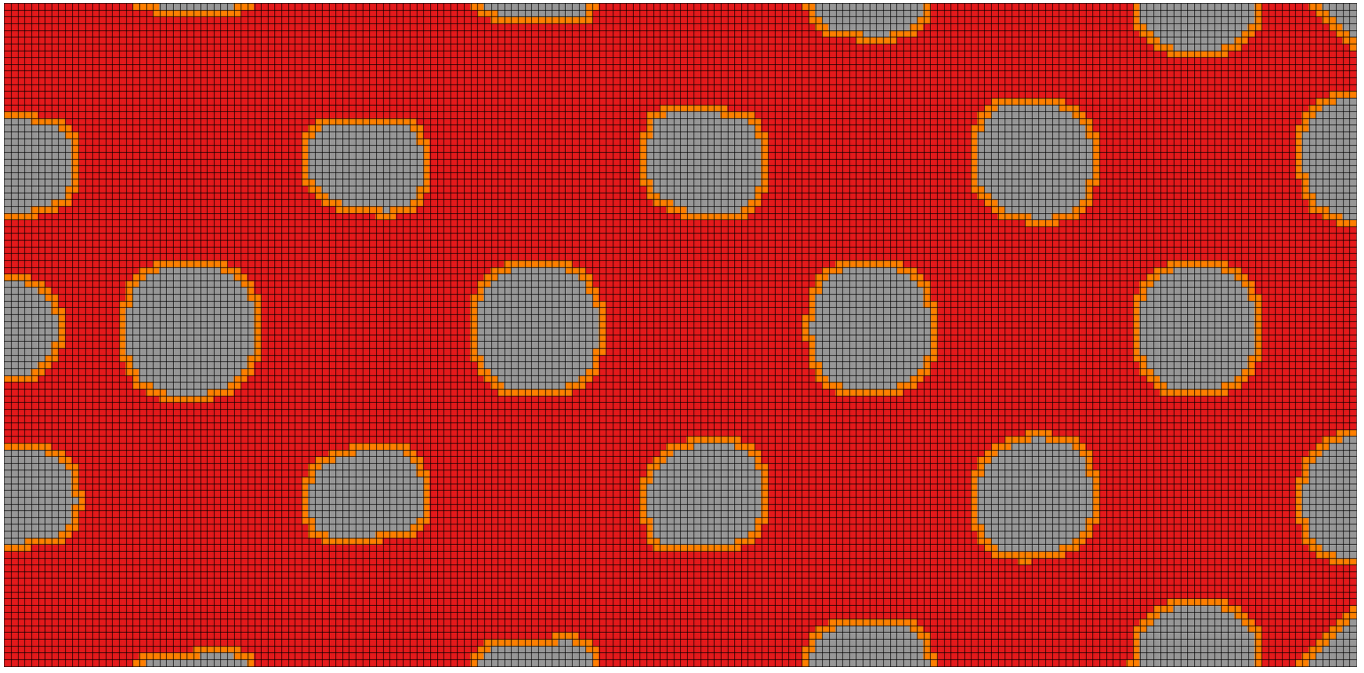
Minimization of the compliance $J(\Omega) := \frac{1}{2}\int_\Omega \sigma(u) : \varepsilon(u) \mathrm{d}x.$

$$\mathrm{d}J_{\Omega_\mathbf{o}}(\theta) = \frac{1}{2}\int_{\Omega_\mathbf{o}} [2(\nabla u)^T \sigma(u) - \sigma(u) : \varepsilon(u) I] : \nabla\theta \mathrm{d}x.$$

See e.g. [Laurain, 2018], [Burman et al., 2018], [Díaz-Avalos, Laurain, 2026].

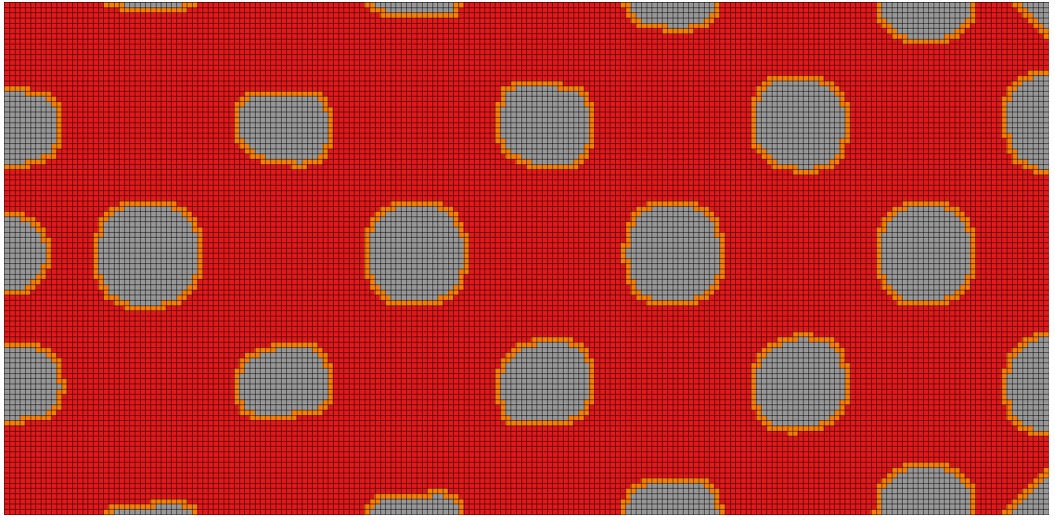
Cantilever beam: φ-FEM formulation (1/3).

φ_h: Lagrange FE interpolation of φ, $\Gamma_h := \{\varphi_h = 0\}$.



Red and Orange : Ω_h | Ω_h^Γ

Cantilever beam: φ-FEM formulation (2/3).



Formally:

⋆ Exterior normal $n = \dfrac{\nabla\varphi}{|\nabla\varphi|}$.

⋆ Auxiliary variables:

- $y := -\sigma(u)$,
- $y\nabla\varphi + p\varphi = 0$ (Ω_h^{Γ}) $\rightarrow$ $yn = 0$ on $\partial\Omega$.

Cantilever beam: φ-FEM formulation (3/3).

Goal: find (u_h, y_h, p_h) s.t. for all (v_h, z_h, q_h),

$$\int_{\Omega_h} \sigma(u_h) : \varepsilon(v_h) \mathrm{d}s + \int_{\partial\Omega_h} (y_h n) \cdot v_h \mathrm{d}s$$

$$+\gamma_{div} \int_{\Omega_h^\Gamma} \mathrm{div}(y_h)\, \mathrm{div}(z_h) \mathrm{d}x + \gamma_u \int_{\Omega_h^\Gamma} (y_h + \sigma(u_h)) : (z_h + \sigma(v_h)) \mathrm{d}x$$

$$+\frac{\gamma_p}{h^2} \int_{\Omega_h^\Gamma} \left(y_h \cdot \nabla \varphi_h + \frac{1}{h} p_h \varphi_h \right) \cdot \left(z_h \cdot \nabla \varphi_h + \frac{1}{h} q_h \varphi_h \right) \mathrm{d}x$$

$$+\sigma h \sum_{F \in \mathcal{F}_{h,\mathbf{1}}^\Gamma} \int_F [\sigma(u_h) n_F][\sigma(v_h) n_F] \mathrm{d}s = \int_{\Gamma_N} g \cdot v_h \mathrm{d}s.$$

"Classical" terms | Stabilization | Penalization | "Ghost penalty"

See [Duprez et al., 2022], [Cotin et al., 2023], [Bulle et al., 2026].

Sources of the multimedia content.

- Slide 3: Laboratoires Serviers (first two pictures) - Wikimedia Commons.
  Links: Picture 1: Artery with plaque ; Picture 2: Artery with balloon.
- Slide 3: [Le Floc'h et al., 2009] (lower right corner of the slide).
  Link: Picture: Artery with plaque (x2).
- Slide 5: GeoGebra (creation from me).
  Link: Picture: Level-set example.
- Slides 10, 18 and 19: R. Bulle.
- Slide 16: M. Duprez (φ-FEM presentation).
  Link: Presentation of φ-FEM.

- Slides 9 and 10: Videos from R. Bulle.
  Links: Video of Slide 9: Gradient descent; Video of Slide 10: Cantilever result.

Other pictures are not referenced (see under the relevant slides).